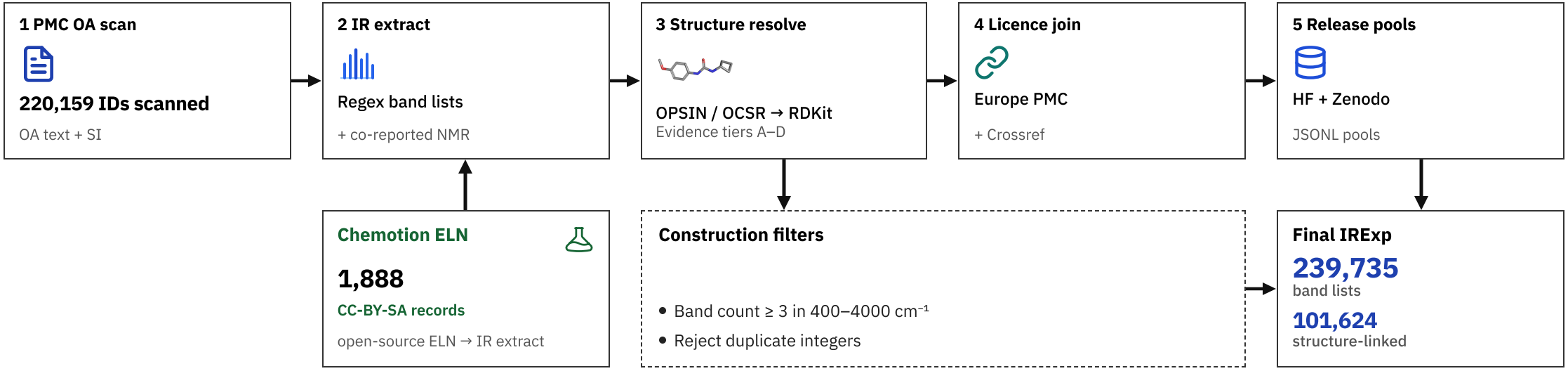
A Extraction and quality-control workflow



B Example automated checks

Band count too low

IR (KBr): `3021` `3021` `2854` cm^{-1} — 2 unique after de-duplication (need ≥3)

IR band outside window

IR (neat): 3200, 2958, `4180` , 1520 cm^{-1} — outside 400–4000 cm^{-1}

^1H integral vs formula

^1H: `(5H)` `(5H)` `(10H)` = `20H` ; $C_9H_{11}NO_2$ (11H) — 20 exceeds H+2 (=13); quarantine flag — row kept